

Heat waves and their significance for a temperate benthic community: A near-natural experimental approach

Christian Pansch¹  | Marco Scotti¹  | Francisco R. Barboza¹ | Balsam Al-Janabi¹ | Janina Brakel¹ | Elizabeta Briski¹  | Björn Bucholz¹ | Markus Franz¹ | Maysa Ito¹ | Filipa Paiva^{1,2} | Mahasweta Saha¹ | Yvonne Sawall³ | Florian Weinberger¹  | Martin Wahl¹

¹Department of Marine Ecology, GEOMAR Helmholtz Centre for Ocean Research Kiel, Kiel, Germany

²MARE – Marine and Environmental Sciences Centre, Quinta do Lorde Marina, Caniçal, Madeira Island, Portugal

³Bermuda Institute of Ocean Sciences (BIOS), St. George's, Bermuda

Correspondence

Christian Pansch, GEOMAR Helmholtz Centre for Ocean Research Kiel, Kiel, Germany.
Email: ch.pansch@gmail.com

Funding information

Alexander von Humboldt Sofja Kovalevskaja Award; Deutsche Bundesstiftung Umwelt; Deutsche Forschungsgemeinschaft; German Federal Ministry of Education and Research; CAPES foundation; Helmholtz-Gemeinschaft; Deutscher Akademischer Austauschdienst

Abstract

Climate change will not only shift environmental means but will also increase the intensity of extreme events, exerting additional stress on ecosystems. While field observations on the ecological consequences of heat waves are emerging, experimental evidence is rare, and lacking at the community level. Using a novel “near-natural” outdoor mesocosms approach, this study tested whether marine summer heat waves have detrimental consequences for macrofauna of a temperate coastal community, and whether sequential heat waves provoke an increase or decrease of sensitivity to thermal stress. Three treatments were applied, defined and characterized through a statistical analysis of 15 years of temperature records from the experimental site: (1) no heat wave, (2) two heat waves in June and July followed by a summer heat wave in August and (3) the summer heat wave only. Overall, 50% of the species showed positive, negative or positive/negative responses in either abundance and/or biomass. We highlight four possible ways in which single species responded to either three subsequent heat waves or one summer heat wave: (1) absence of a response (tolerance, 50% of species), (2) negative accumulative effects by three subsequent heat waves (tellinid bivalve), (3) buffering by proceeding heat waves due to acclimation and/or shifts in phenology (spionid polychaete) and (4) an accumulative positive effect by subsequent heat waves (amphipod). The differential responses to single or sequential heat waves at the species level entailed shifts at the community level. Community-level differences between single and triple heat waves were more pronounced than those between regimes with vs. without heat waves. Detritivory was reduced by the single heat wave while suspension feeding was less common in the triple heat wave regime. Critical extreme events occur already today and will occur more frequently in a changing climate, thus, leading to detrimental impacts on coastal marine systems.

KEYWORDS

Baltic Sea, benthic temperate community, climate extremes, community experiment, marine heat waves, near-natural experiment

1 | INTRODUCTION

Global climate models project warming and acidification of the oceans worldwide (IPCC, 2013). At a regional scale, for the Baltic Sea, freshening of seawater, intensification of hypoxia, and increased eutrophication are projected for the near future (Gräwe, Friedland, & Burchard, 2013). Superimposed on these long-term trends is an increase in the variability around mean changes, particularly in the frequency, intensity and the duration of extreme events (Rahmstorf & Coumou, 2012). Shifts of the means but also of the variability of environmental factors will certainly modify key biological processes and alter structure and functioning of ecosystems (Benedetti-Cecchi, Bertocci, Vaselli, & Maggi, 2006; Wernberg, Smale, & Thomsen, 2012).

Terrestrial ecologists have started to acknowledge the effects of environmental variability on species' performance, dynamics and distribution, and the consequences on community structure, ecosystem functioning and services (e.g. Bozinovic et al., 2011; Estay, Clavijo-Baquet, Lima, & Bozinovic, 2011; Jentsch, Kreyling, & Beierkuhnlein, 2007; and references therein). Recent analyses on temperature performance curves of terrestrial ectothermic invertebrates indicated that increased temperature variation poses a greater risk to species than mean warming, while in combination these factors represent the greatest risk (Paaajmans et al., 2013; Vasseur et al., 2014). In the marine domain, knowledge on the expected ecosystem changes in response to environmental variability is less advanced. Here, environmental fluctuations, which are an integral part of nature, have often been excluded from experimental designs for decades (Boyd et al., 2016; Helmuth et al., 2014; Wahl, Saderne, & Sawall, 2016), following the old paradigm that the effect of a driver can only be identified if the manipulated levels of this driver, and other experimental conditions, are kept constant. Besides this, an upscaling from single species to ecosystems, from single to multiple drivers, and from short-term incubations to long-term adapted species and communities, is urgently needed (Riebesell & Gattuso, 2015).

One example of environmental fluctuations are heat waves, as recently defined by Hobday et al. (2016). Field observations have demonstrated strong responses of marine environments to heat waves, from single species to the ecosystem level (Wernberg et al., 2013). To date, however, the ecological effects of marine heat waves are mainly deduced from correlations between naturally occurring temperature anomalies and concurring species or community shifts (Daufresne, Bady, & Fruget, 2007; Garrabou et al., 2009; Glynn & Dcroz, 1990; Harris et al., 2007; Johnk et al., 2008; Reusch, Ehlers, Hammerli, & Worm, 2005; Swaty, Deckert, Whitham, & Gehring, 2004; Wernberg et al., 2013). Moreover, the impacts of heat waves have been rarely studied in temperate regions. Controlled experimental studies—in which most other factors but the exerted extreme thermal event can be excluded—are almost entirely lacking, particularly so at the ecosystem level (but see Sorte, Fuller, & Bracken, 2010). Recent improvements in mesocosm technology allow multiple-species experiments with multifactorial designs, including simulated heat waves (Stewart et al., 2013; Wahl et al., 2015).

Heat waves, in contrast to the global warming driven increase in annual mean temperature, represent discrete events of potentially high thermal stress. Subsequent heat waves are separated by phases of low(er) stress during which organisms may recover. Indeed, occasional anomalies in temperature of stressful but sublethal intensity and duration may “strengthen” a population (Woodin, Hilbish, Helmuth, Jones, & Wetthey, 2013). Sequential stressful events with intermittent stress relaxation may have cumulated additive, synergistic or antagonistic effects (Gunderson, Armstrong, & Stillman, 2016). Which of these interactions materialize depends on species identity, stressor identity, the existence of stress memory and the duration of the recovery phase (Gunderson et al., 2016; Walter, Jentsch, Beierkuhnlein, & Kreyling, 2013).

Coastal benthic ecosystems are among the world's most productive ecosystems (Costanza et al., 1997), providing vital ecosystem services such as food production, energy transfer in food webs, biogeochemical cycling, water clearance and provisioning of nursery areas for higher trophic-level taxa (Griffiths et al., 2017 and references therein). At the same time, they experience particularly strong anthropogenic pressure from dense human populations, which mainly occur in coastal areas (Halpern et al., 2008; Lotze et al., 2006). Also the proximity of land masses and their shallow depth make coastal ecosystems more vulnerable to extreme events such as heat waves (Cloern et al., 2016; Halpern et al., 2008). Low functional redundancy, as expected in the species-poor Baltic Sea, may increase ecosystem vulnerability (Micheli & Halpern, 2005).

In the present experimental study, we investigated whether summer heat waves have detrimental consequences on the macrofauna of a temperate coastal community in the Baltic Sea and whether sequential heat waves provoke an increase or decrease in sensitivity to thermal stress. The experiment was carried out in the Kiel Outdoor Benthocosms (Wahl et al., 2015) and consisted of three treatments: 0HW = no heat wave; 1HW = one single summer heat wave; and 3HW = three subsequent heat waves. In all three treatments, working with continuous flow-through of fjord water maintained the natural small-scale variability of most environmental parameters except temperature, which was adjusted to the logged thermal regime of the year 2009 (see details below). To the 2009 regime, programmed heat waves were or were not added. Based on previous experiments (Werner, Graiff, & Matthiessen, 2016), we postulated that summer heat waves would be detrimental to the performance or even the survival of some species. We expected that the occurrence of earlier heat waves in the year and, thus, at lower mean temperatures, might affect the impact of the summer heat wave, i.e. “strengthen” or preparing species to thermal stress (Woodin et al., 2013).

2 | MATERIALS AND METHODS

2.1 | Experimental setup

The experiment was conducted between May 6 and August 20, 2015 in the Kiel Outdoor Benthocosms (KOBs; see a detailed

description of the setup in Wahl et al. (2015)). The KOB is a permanent experimental facility floating alongside a jetty in the Kiel Fjord, Germany (54°19'47.74"N, 10°8'58.14"E). It consists of six thermally insulated large mesocosms, each divided into two independent experimental units of 1,500 L. In the present experiment, a constant flow-through of 1,800 L per day of unfiltered seawater allowed natural fluctuations in both abiotic (salinity, pH, oxygen and nutrients) and biotic (bacterial load, phytoplankton and zooplankton) conditions, all driven by seasonality, weather and hydrography (Wahl et al., 2015). Only temperature was directly controlled. The system was exposed to natural light conditions throughout the experiment.

2.2 | Control treatment

A generalized additive mixed model (GAMM) was fitted to the 15-year time series data set (2000–2014) of daily mean temperatures registered in the Kiel Fjord at 1.5 m depth, representative of Baltic Sea shallow water communities (54°19'45.97"N, 10°8'58.58"E). Data were provided by the Marine Meteorology research unit of GEOMAR (F Nevoigt, U Hecht and K Bumke; www.geomar.de/en/service/weather). The GAMM was applied for identifying long-term changes at the overall time series level and within-year variation (i.e. seasonal term). The analysis was performed using the function "gamm" from the "mgcv" R package (Wood, 2004, 2006). Dimensions of the basis used for the smooth term were set to the maximum possible value for the months ($k = 12$), and a cyclic cubic spline was adopted to prevent discontinuity between January and December in the seasonal term. Residual autocorrelation was tested with the (partial) autocorrelation function. Autoregressive moving average (ARMA) processes were applied to the residuals for assembling candidate models with correlated errors. The candidate models were constructed with three possible orders ($p = 1, 2, 3$) for the autoregressive component (AR), and the ARMA was nested within each month. A generalized likelihood ratio test was applied to decide which of the candidate models fitted the data best. The smooth term of the seasonal variation was used to identify the temperature of the control treatment. In order to preserve natural stochastic fluctuations in the control treatment, the temperature profile of the year with the lowest deviation from the seasonal term modelled with the GAMM (at 95% confidence interval) was selected as control, i.e. the year 2009 from the tested data set. For each year, the deviation was measured as number of days with temperature outside the limits of the 95% confidence interval. Day/night fluctuations were included in the control treatment for reproducing a more realistic temperature profile. The experimental units attained highest daily temperature at midnight and lowest daily temperature in the morning (at 9 a.m.). The choice of these hours reflects the frequency distribution of hourly minimum and maximum temperatures from May 1 to September 30, 2009 (Figure S1; Data S1). The daily extreme temperatures were obtained using hourly values of Kiel Fjord at 1.5 m water depth in 2009. Linear interpolation was implemented for temperature changes between daily extremes.

2.3 | Heat wave treatments

To design the heat wave treatments, daily changes in mean temperatures of the Kiel Fjord at 1.5 m depth from May 1 to September 30 (years 2000 to 2014) were considered (i.e. the season the current experiment ran). Heat waves were described as events during which the daily temperature increases over two or more consecutive days exceeding a given threshold. Such threshold was defined as the highest daily increase that allowed finding, at least, one heat wave in each year of the time series. In the period 2000–2014, this threshold corresponded to 0.7°C/day. This threshold was used to identify both heat waves and strong cooling events (descriptive statistics in Tables S1–S6). The amplitude adopted for the first two heat waves was such that the 3HW treatment attained temperatures 3.6°C higher than the control (this value is the maximum of the yearly average amplitudes; Table S2, Figure S2). The heating lasted three days at a rate of 1.2°C/day (i.e. the highest average daily increase recorded on a yearly basis; Table S3). For a period of four days, the 3HW treatment was kept at 3.6°C warmer than the 0HW treatment (i.e. the median value of the days that pass between heat waves longer than two days and subsequent strong cooling phases; Table S7). The cooling phase lasted two days (i.e. median number of days with a cooling rate $< -0.7^\circ\text{C}/\text{day}$; Table S4), and the rate depended on the difference between the 3HW treatment and the control (0HW). The cooling rates were always included in the realistic boundaries defined by minimum and maximum average daily decrease ($-2.0^\circ\text{C}/\text{day}$ and $-0.8^\circ\text{C}/\text{day}$; Table S6). The structure of the August heat wave implemented for both heat wave treatments (1HW and 3HW) followed the pattern of the previous heat waves, i.e. 3 days of warming, 4 days of peak phase during which the difference between heat wave treatments and control was preserved, and 2 days of cooling. Here, however, the intensity of the heat wave was increased with an amplitude of 5.2°C (i.e. average of the four most extreme heat waves in the period 2000–2014) and a warming rate of about 1.7°C/day, to maintain the structure of 3 days of warming. This value is realistic, being lower than the maximum rate found in the time series: 2.6°C/day (see Table S3). Day/night fluctuations were implemented for both heat wave treatments. The intensity and the direction of day/night fluctuations (even during heat wave events) followed the same pattern as that of the control treatment.

The maximum temperatures implemented were 20.4°C for the 0HW treatment, and 25.2°C for the 1HW and 3HW treatments (Figure 1).

2.4 | Achieved treatments

Over the duration of the experiment, temperature was logged with a 10-min time interval by GHL sensors (platinum resistance thermometer PT1000, from GHL Advanced Technology, Kaiserslautern, Germany). For more details on the computer system and the data processing, see Wahl et al. (2015). Temperature treatments followed the programmed temperature profile, with and without modelled heat waves (Figure 1). Organisms were kept at Fjord temperature in

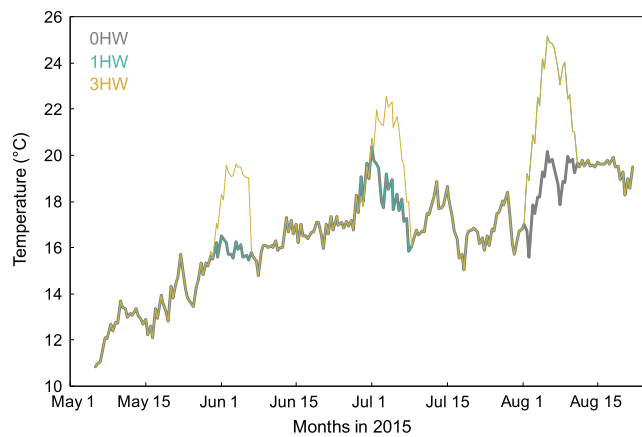


FIGURE 1 Implemented temperature profile within the benthocosm units. Temperatures are provided from May to August 2015 (experimental period) for the 0HW (grey), the 1HW (green) and 3HW (yellow) treatment

the benthocosms for 4 days before this temperature was slowly brought in line with the ambient treatment temperature (i.e. 0HW, 2009 temperature; Figure S3). Thereafter, deviations from programmed values were only recorded during some exceptionally warm days, mostly during late afternoon (confidence intervals in Figure S3). The temperature profile was adjusted and deviated from the 2009 profile by about 2°C during six extremely warm days in July (see period July 7–12 in Figures 1 and S3).

The maximum temperatures of 25.2°C for both heat wave treatments (Figures 1, S3) exceeded maximum temperatures that occurred over the past 15 years in shallow, open waters of the Kiel Fjord by about 1°C (Figures S3, S4).

Seawater salinity, pH, oxygen and nutrients were, in addition to temperature, recorded during the experiment in all benthocosm tanks and the Kiel Fjord. Salinity as measured by a handheld salinity meter (WTW Cond 3110, TetraCon 925, Weilheim, Germany) ranged between 13.9 and 16.4 with a mean salinity of 15.4. Fjord mean salinity was 15.2. Salinity data are provided within Pangaea, pH, oxygen and nutrients will be presented in line with another manuscript.

2.5 | Experimental community

Organisms were collected within the Kiel Fjord between May 4 and 6, 2015, at a subtidal *Zostera marina* belt (Falckenstein: 54°24'24.69" N, 10°11'38.74"E; *Z. marina*, *Cerastoderma edule*), in a shallow intertidal zone (Bülk: 54°27'8.13"N, 10°11'58.42"E; *Fucus vesiculosus*, *Gammarus salinus*, *Gammarus locusta*, *Idotea baltica*, *Littorina littorea*) as well as directly at the GEOMAR pier (54°19'45.97"N, 10°8'58.58" E; *Mytilus edulis*). Here, the two main habitat forming macrophytes (*Z. marina* and *F. vesiculosus*) as well as their most abundant associated faunal components, representative of Western Baltic Sea shallow water communities, were chosen and added in natural abundances (see Werner et al., 2016). Organisms were either planted (or placed) in beakers with sediment (and then arranged in the experimental units) or directly released to the experimental units. Sediment

was collected from the *Z. marina* belt using a Van Veen grab and sieved (2,000 µm) before being used in the experiment. *Zostera marina* (86 individuals) and *F. vesiculosus* (five individuals) were placed into each experimental unit providing habitat, shelter and food for benthic macrofaunal species. Macrophytes were distributed inside the units using 1 L aquaria and 16 beakers with 3.0 and 0.8 L of sediment, respectively. The performance of macrophyte species *Z. marina* and *F. vesiculosus* is not included in the present manuscript. In addition, 15 beakers, containing 1.4 L of sediment with four *C. edule* individuals, were placed inside the units. Since *C. edule* were regularly removed from the experimental units for evaluating their physiology in a parallel project, this species was not included in the final analysis. The sediment offered habitat for the recruitment of sessile/infauna species from the flow-through of unfiltered Kiel Fjord water. The amphipods *G. salinus* and *G. locusta* (50 individuals each), the isopod *I. baltica* (20 individuals), the mussel *Mytilus edulis* (six individuals) and the snail *Littorina littorea* (80 individuals) were added and left to move freely in the experimental units. All species were added at equal numbers to each experimental unit, where these numbers corresponded to the species prevalence typically observed in coastal communities of the Kiel Fjord (Werner et al., 2016).

2.6 | Determination of abundance and biomass

At the end of the experiment, sediment infauna was sampled from five random 1.4 L beakers of each experimental unit (sessile/infauna). The sediment of the beakers was pooled, sieved (1,000 µm) and fixed in 37% formalin. Collected organisms were identified to the lowest possible taxonomic level, dried (80°C for 24 hr) and weighed (Sartorius, Göttingen, Germany; 0.1 mg). Species abundance and biomass quantified in the sieved subsamples were extrapolated to the overall volume of sediment included in the experimental unit (7 L of sampled sediment and 67 L of total sediment per experimental units). Noninfauna organisms (free-living) were sieved (1,000 µm), collected in bags and frozen at the end of experiment. As in the case of sessile/infauna organisms, collected animals were sorted and identified to the lowest possible taxonomic level. Abundance and size of sorted animals was obtained using ZooScan (Gorsky et al., 2010). The biomass was calculated from size-to-dry weight relationships (Delgado, Guerao, & Ribera, 2011; Rumohr, Brey, & Ankar, 1987; Weinberger et al., 2011). Data refer to the number of individuals or biomass per experimental units (volume of 1500 L, bottom and side-wall area of 5.6 m²).

2.7 | Statistics

The effects of the 1HW and 3HW treatments on the abundance and biomass of 14 macrofaunal (sessile/infauna and free-living) species were evaluated through generalized linear mixed models (GLMM) performed with the "lme4" R package (Bates, Maechler, Bolker, & Walker, 2017). For this analysis, species were only considered if they occurred in at least three of the 12 experimental units and exhibited abundances higher than one individual in the sediment

sampled of at least one experimental unit. For each species, a GLMM was run including abundance or biomass as response variables and treatment as explanatory variable (fixed factor). Identities of benthocosm pairs were included in the model as random intercepts, adjusting potential sources of variability related to the position of the experimental unit on the KOB platform (Wahl et al., 2015) and to model the dependency between the measurements obtained in the two half units of the same mesocosm. The negative binomial and gamma distributions were used to model abundance (discrete variable) and biomass (continuous and strictly positive variable) respectively. For all models, a logarithmic link function was used. In the frame of the adjusted GLMM, the relative effects of the 1HW and the 3HW treatments in relation to the 0HW treatment were expressed as logarithmic response ratios (from now on referred to as response ratios), using the Wald test to determine if they were significantly different from zero. The adequacy of all adjusted models was evaluated by checking the plots of residuals.

Community shifts based on biomass of all identified species were analysed using the PCO module in Primer 7 (Primer-W Ltd., Plymouth) followed by a PERMANOVA based on Bray–Curtis similarity resemblance matrices after square-root transformation of the raw data. Two sets of analyses were run, one based on the biomasses of single species, the second on the biomasses of functional groups (see Tables S8 and S9 for identified species and functional traits; Wahl, 2009).

3 | RESULTS

3.1 | Species abundance and biomass

All abundance as well as biomass data are presented in Table S10. When compared to the 0HW treatment, 1HW significantly reduced the abundance and biomass of the spionid polychaete *Polydora cornuta* (by 62% and 98%, respectively) and the abundance of the blue mussel *Mytilus edulis* (by 45%; Figure 2a, b). In contrast, the 1HW treatment significantly increased the biomasses of *Marenzelleria viridis* and the grazing amphipod *Gammarus locusta* (by 42% and 33% respectively; Figure 2b). The 3HW treatment significantly reduced the biomass of *P. cornuta* (by 90%) as well as abundance and biomass of the Baltic clam *Limecola balthica* (by 67% and 80% respectively), but significantly increased the biomasses of the amphipod *Corophium* sp. and the polychaete *Nereis* sp. (by 288% and 103% respectively; Figure 2).

Overall, 50% (seven out of 14) of the species showed positive, negative or positive/negative responses to either of the applied heat wave treatments, in either abundance and/or biomass (Figure 2). While six out of seven species (86%) considered as sessile/infauna (SI) showed an effect of the applied treatments, only one of seven free-living (FL) species (14%) were affected (see Figure 2, also for indication of SI and FL).

Heat waves affected four species positively and four species negatively in terms of increased or reduced abundance and/or biomass. Comparing the heat wave impact between the naive (1HW)

and the conditioned (3HW) populations, we observe the tendency of 3HW to be less detrimental than a single summer heatwave: In seven of the eight cases where the different heat wave treatments (1HW and 3HW) had differing impacts (Figure 2).

In summary, half of the investigated species were impacted by the applied heat waves. The bivalve *L. balthica* showed a cumulative stress by the three subsequent heat waves, whereas the polychaete *P. cornuta* showed some sign of a “strengthening” effect by repetitive heat waves (at least with respect to abundances). The sessile amphipod *Corophium* sp. benefitted from the three heat waves.

3.2 | Community analysis

Community shifts, based on species biomasses are presented in Figure 3a. Different responses of species to the heat wave treatment should lead to structural, and perhaps, functional shifts in the communities. The structural changes of the community in response to 1HW or 3HW were not significant (ANOSIM pairwise test: 0HW vs. 1HW, $p = .11$; 0HW vs. 3HW, $p = .31$). The structural differences between communities exposed to 3HW and 1HW, respectively, were marginally significant (ANOSIM pairwise test: 3HW vs. 1HW, $p = .057$). This difference was mainly driven by (in decreasing order of contribution): *Marenzelleria viridis* (12%), *Nereis* sp. (10%), *Littorina littorea* (9.5%), *Corbula gibba* (9%), *Idotea balthica* (6.5) and *Limecola balthica* (6.5%), which cumulatively contributed 51% to the overall differences. Relative to the 3HW regime, 1HW tended to favour *L. balthica* (4 times more abundant), *M. viridis* (+50%) and *Nereis* sp. (+6%) but disfavoured *L. littorea* (−50%), *C. gibba* (absent under 1HW) and *I. balthica* (−35%).

Community shifts, based on functional traits are presented in Figure 3b. The overall functional shift driven by heat waves was significant (ANOSIM: $p = .031$). Heat waves in general favoured crawling or burrowing predators and suspension feeders, while the abundance of detritivores decreased. The pairwise differences in functions were, however, only marginally different between 1HW and 3HW ($p = .057$). Detritivores became about 3 times less common in the 1HW treatment. The locomotory trait burrowing contributed 12% to the overall difference and was higher by 44% under 1HW compared to 3HW. The trophic trait suspension feeding contributed 10% to the overall functional difference and was substantially more common under 1HW. These five traits cumulatively contributed 50% to the functional shift from 3HW to 1HW.

4 | DISCUSSION

In the literature, different approaches have been suggested to define marine heat waves (Marbà, Jorda, Agusti, Girard, & Duarte, 2015; Meehl & Tebaldi, 2004; Meehl et al., 2000; Scannell, Pershing, Alexander, Thomas, & Mills, 2016; Sorte et al., 2010). In one of the most convincing approaches, Hobday et al. (2016) defined heat waves as a period lasting “for five or more days, with temperatures warmer than the 90th percentile based on a 30-year historical

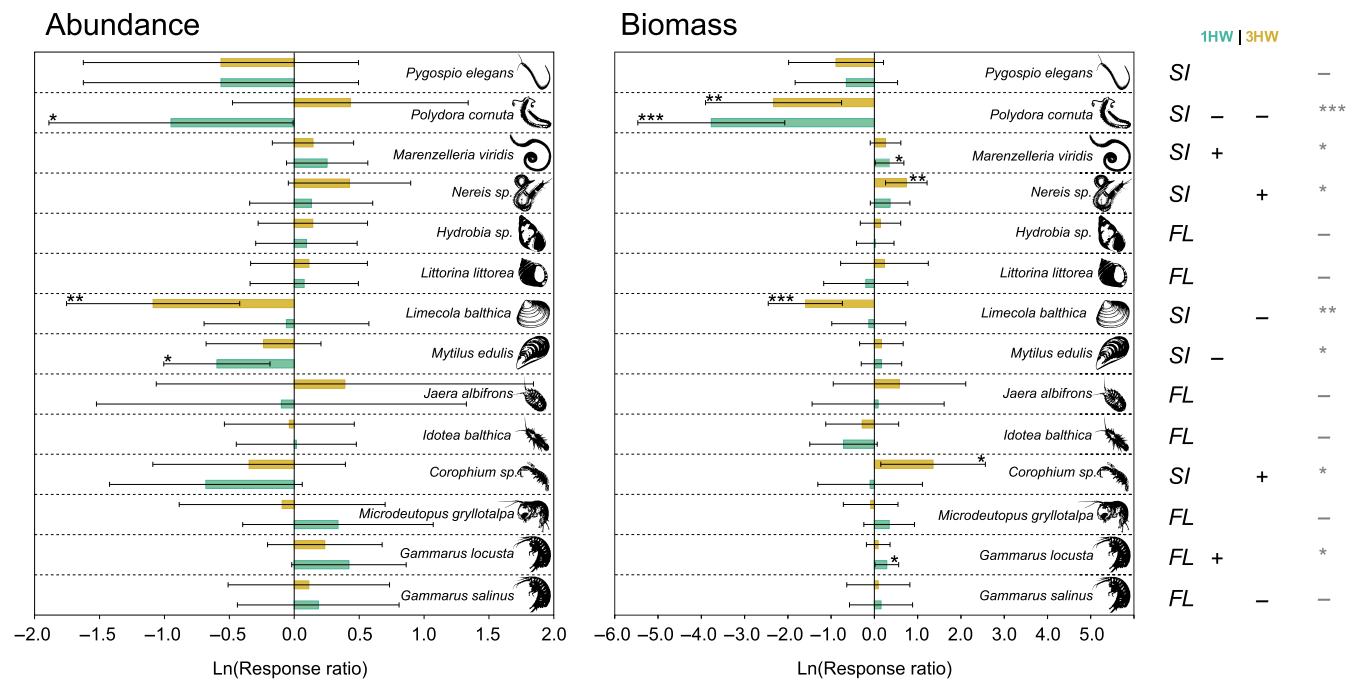


FIGURE 2 Logarithmic response ratios for the abundance and biomass of 14 macrofaunal species. Data show the proportional change in the means of 1HW (green) and 3HW (yellow) treatments with respect to the control (0HW). Response ratios and 95% confidence intervals were calculated based on the estimates obtained from negative binomial and gamma generalized linear mixed models (GLMM) adjusted for abundance and biomass, respectively (see details in the main text). Asterisks show response ratios significantly different from zero (* $p < .05$, ** $p < .01$ and *** $p < .001$). Species were classified into sessile/infauna (SI) and free-living (FL) species. Positive and negative effects of applied treatment are summarized by “+” and “−”, respectively

baseline period”. This definition will certainly help detecting marine heat waves in nature, as a basis for monitoring their impacts on community structure and functioning (Garrabou et al., 2009; Jentsch et al., 2007; Wernberg et al., 2013). Experimental assessments, however, are needed for better understanding the mechanisms underlying the impacts of marine heat waves observed in the field (Sorte et al., 2010), one of the few experimental community studies on marine heat waves, applied dramatic heating (3°C/h) and cooling rates (2°C/h) to simulate heat wave events. In the present study, we applied a more realistic approach by testing the community-level effects of marine heat waves in treatments in which parameters were extracted from natural temperature profiles grafted onto a seasonally, diurnally and stochastically fluctuating temperature regime of the Kiel Fjord in 2009.

Applied heat wave treatments affected, beneficially or detrimentally, 50% of 14 considered macrofauna species. Maximum temperatures applied in both heat wave treatments (25.2°C) exceeded maximum temperatures that sporadically occurred over the past 15 years in shallow waters of the Kiel Fjord (24°C; Figure S4) only marginally. Thus, the “absence” of an effect (50% of the species, Figure 2) can be interpreted as a long-term acclimation of Baltic Sea populations to a thermally variable shallow sea. Nevertheless, in single species (*Polydora cornuta*, *Marenzelleria viridis*, *Nereis sp.*, *Limecola balthica*, *Mytilus edulis*, *Corophium sp.* and *Gammarus locusta*; Figure 2), we identified distinctive responses to the applied heat wave treatments. Species-specific effects were also observed by Sorte

et al. (2010) who worked on short-term heat wave impacts on marine epibenthic fouling communities of Bodega Harbour, California, United States (see also Mouthon & Daufresne, 2006; Parmesan, Root, & Willig, 2000 for terrestrial and freshwater systems, and Harley, 2008 for marine intertidal areas). Thus, heat waves have the capacity to affect community structure via shifts in species abundance and/or biomass up to complete eradication of single species from the community (Allison, 2004). Since we worked with entire communities, the observed species-level effects may also be an indirect response mediated by shifts in biotic interactions, i.e. “post-interactive” effect (*sensu* Woodin et al., 2013).

A variety of response patterns was observed. The tellinid bivalve *Limecola (Macoma) balthica* was negatively impacted by the three subsequent heat waves, while the single summer heat wave did not significantly impact abundance or biomass of this species. This indicates a possible “additive” negative effect from frequently occurring heat waves due to an accumulation of temperature stress (Gundersen et al., 2016; Sokolova, Frederich, Bagwe, Lannig, & Sukhotin, 2012), and the absence of an ecological stress memory as described by Walter et al. (2013). *Limecola balthica* can be a dominant species in shallow soft-bottom habitats worldwide (Beukema, Dekker, & Jansen, 2009; Holland, Mountford, & Mihursky, 1977). In the more brackish Baltic Sea, *L. balthica* is regarded as a key species, often dominating the biomass in soft sediments (Bonsdorff, Norkko, & Boström, 1995; Laine, 2003). This species escapes hypoxia by upwards migration to shallower burial depths and siphon extension

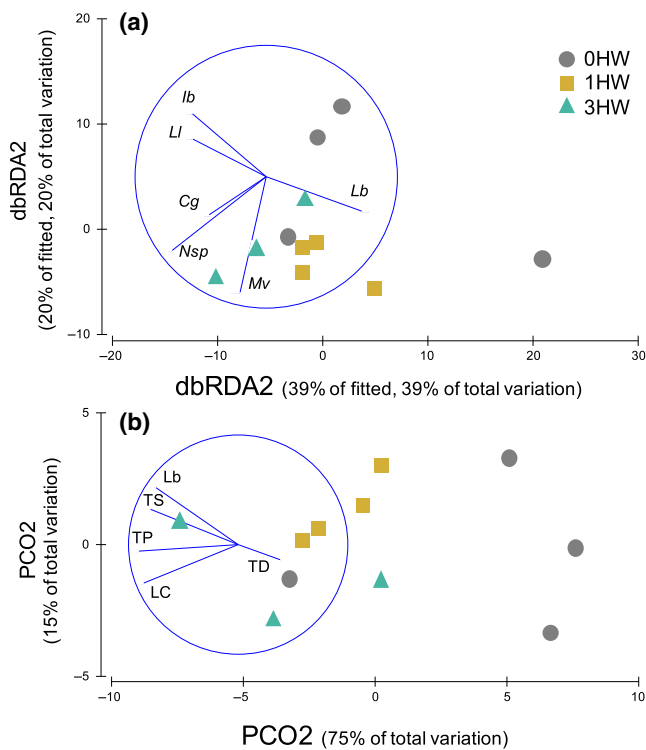


FIGURE 3 Dissimilarities among communities. Data were analysed using the PCO module in Primer-PERMANOVA representing dissimilarities among differently treated communities based on biomasses of species (a) and of functional traits of species (b). The overlays illustrate how biomass of species or functional traits shift, which cumulatively contribute 50% to the community shifts under the various heat wave treatments (1HW vs. 3HW). Main species: *Idotea balthica* (*Ib*), *Littorina littorea* (*LI*), *Corbula gibba* (*Cg*), *Nereis* sp. (*Nsp*), *Marenzelleria viridis* (*Mv*) and *Limecola balthica* (*Lb*). Main functional traits: burrower (*LB*), suspension feeder (*TS*), predator (*TP*), crawler (*LC*), detritivore (*TD*)

(Long, Seitz, Brylawski, & Lipcius, 2014 and references therein). Its tolerance to warming oceans, however, seems low. Indeed, this species was shown to be more sensitive to warming, than many other bivalve species of the same habitat (e.g. *Mya arenaria*, *Cerastoderma edule* and *Mytilus edulis*; Van Der Veer, Cardoso, & Van Der Meer, 2006). A gradual decline of *L. balthica* populations over the past decades was linked to the occurrence of warmer than usual summer temperatures (Beukema et al., 2009). Most of these studies, however, concentrated on the increase of temperature means. The current study demonstrates that intermittent warming (by the applied heat waves) can also strongly affect this species.

In contrast to *L. balthica*, the spionid polychaete worm *P. cornuta* (formerly *P. ligni*; Blake & Maciolek, 1987) declined in abundance and biomass more strongly in response to the single summer heat wave than to the triple heat wave. This species is common in intertidal and shallow subtidal habitats worldwide (Dauer, Maybury, & Ewing, 1981; Zajac & Whitlatch, 1982). *Polydora cornuta* is an "interface" feeder switching between suspension and deposit feeding depending on flow speed and food resources (Bock & Miller, 1996; Dauer et al., 1981; Taghon, Nowell, & Jumars, 1980). In apparent

contrast to the observed sensitivity to the applied summer heat wave, this species has been reported to be tolerant to a wide range of salinities and temperatures (Dauer et al., 1981; Shimeta, Witucki, & Hippe, 2004; Zajac, 1991a,b). The fact that *P. cornuta* was less impacted by the three consecutive heat waves than by the single summer heat wave, might be indicative of "acclimation" (*sensu* Ainsworth et al., 2016; Gunderson et al., 2016), or ecological stress memory (*sensu* Walter et al., 2013). In addition, under triple heat waves, in contrast to one single heat wave, the population tended to increase in abundance but decrease in biomass, which might indicate enhanced reproduction. If only the more robust offspring survive, this selection could enhance thermal robustness over time. *P. cornuta* is one of the most successful invasive species in the Mediterranean region (Zenetos et al., 2012) and worldwide (Dauer et al., 1981; Zajac, 1991a,b). Due to its short generation time, high reproductive output and substantial juvenile mortality, *P. cornuta* is capable to adapt to new environmental conditions rapidly (e.g. Zajac, 1991a). Thus, a sequence of moderately stressful heat waves could "train" this species for future warm conditions.

Three subsequent heat waves increased the biomass of the amphipod *Corophium* sp. considerably, when compared to the treatment without a heat wave. The genus *Corophium* is generally considered to be an unselective deposit feeder, although suspension feeding may occur (Gerdol & Hughes, 1994a,b). *Corophium volutator* for example, can be extremely abundant (up to 100,000 individuals per m²) with strong impacts on diatom abundance (Gerdol & Hughes, 1994b) and species composition (Hagerthey, Defew, & Paterson, 2002). The optimum temperature of this amphipod was found to be around 18°C (Meadows & Ruagh, 1981). The optimum temperature for *Gammarus* sp. in the Western Baltic seems to be around 21°C, beyond which mortality rapidly increases (Gülzow, 2015). Below this threshold temperature, however, warming enhances reproduction of *Gammarus* sp. (Werner et al., 2016). As with *P. cornuta* (see above), short generation time of *Corophium* (Skadsheim, 1990) is generally assumed to facilitate adaptation (but see Johannesson & Andre, 2006) to, e.g. a warmer climate under a regime of successive heat waves.

Faced with stressful (transient) conditions, organisms can adopt different mechanisms: (1) temporal avoidance of habitats, (2) evolution of stress resistance (increased plasticity) and/or (3) if damaged, activation of recovery mechanisms (Huey et al., 2002). In the present study, 86% of the species considered as sessile/infauna responded to the applied heat wave treatments, while only 14% of the free-living species were affected. In nature, mobile species can avoid shallow heated areas by migrating into deeper and colder water layers (Huey et al., 2002; Schild & Glauser, 2013), particularly so in strongly stratified water bodies as is the Kiel Fjord in summer. However, these potential thermal refugia in deeper water tend to become acidic and hypoxic in late summer (Melzner et al., 2012), and thus, hostile to most species. In the present study, temperature within single experimental units was equal throughout the tanks (Wahl et al., 2015), and thus, heat avoidance can be excluded as an explanation for our results. The gastropod *Littorina littorea* was the only species

capable of temporally avoiding the heated water body during colder nights (by leaving the water phase), and could, by adjustment of body temperatures through a range of physiological or morphological adaptations, have overcome the heat stress (Chappon & Seuront, 2011 and references therein). For the time being, we thus have no good explanation why mobile species were thermally more robust.

The differential responses to single or sequential heat waves at the species level entailed shifts at the community level. The relative contribution of single species to the shifts at the community level, whether taxonomic or functional composition was considered, did also depend on the dominance of a species and, thus, may not scale with the impact strength observed at the species level. Community shifts were less significant (ANOSIM $p = 6.7$) when considering species biomass shifts than when analysing the shifts in functional trait prevalence (ANOSIM $p = .03$). Surprisingly, the community-level differences between single and triple heat waves were generally more pronounced than those between regimes with vs. without heat waves. The significant functional shifts were mainly due to the (marginally significant) differences between 1HW and 3HW regimes. Detritivory was reduced by the single heat wave while suspension feeding was less common in the triple heat wave regime. Overall, however, the treatment driven shifts were not particularly pronounced (15%–20% and 5%–9% dissimilarities regarding species and functional traits, respectively). This, in line with a recent experimental study and meta-analysis, suggests that single species effects from climate drivers may be buffered by compensatory dynamics in more complex community assemblies (Goldenberg et al., 2018).

In the present study, we applied a new approach for experimentally testing the effects of marine heat waves, i.e. by the inclusion of natural seasonality as well as stochastic and diurnal variability. We found that heat waves can have detrimental as well as positive effects on single compartments of a temperate benthic community of the western Baltic Sea. We highlight four possible ways in which single species responded to either three subsequent heat waves or one single summer heat wave: (1) the absence of a response (tolerance), (2) negative accumulative effects by three subsequent heat waves, (3) buffering by proceeding heat waves due to acclimation and/or shifts in phenology and (4) accumulative positive effect by subsequent heat waves. Interestingly, the majority of motile species (free-living component of the species) in the system did not respond to heat waves, while infaunal species were mostly affected. The responses to single or sequential heat waves at the species level entailed shifts at the community level, with likely consequences for marine temperate ecosystems, such as the Baltic Sea. As critical thresholds such as maximum temperature, amplitude or duration of heat waves will occur more frequently in a changing climate, changes in the structure and functioning of coastal marine ecosystems can be expected.

DATA AVAILABILITY STATEMENT

All data are available at PANGAEA database: <https://doi.org/10.1594/pangaea.888578>. (Pansch et al. 2018).

ACKNOWLEDGEMENTS

This project was partly funded by Helmholtz and by GEOMAR as well as by the Cluster of Excellence “The Future Ocean”. The “Future Ocean” is funded within the framework of the Excellence Initiative by the Deutsche Forschungsgemeinschaft (DFG) on behalf of the German federal and state governments. FRB acknowledges the financial support of the German Academic Exchange Service (DAAD) through the project Doctoral Programmes in Germany 2015/16 (57129429). MI acknowledges financial support of CAPES foundation (Ministry of Education of Brazil) through the Doctoral Programme (process number: 99999.001303/2015-05). MS acknowledges funding from the Cluster of Excellence project CP1215 through a Post-Doctoral Research Grant. BA acknowledges financial support from the Project BIOACID II of the German Federal Ministry of Education and Research (BMBF; FKZ 03F0655, A). JB acknowledges financial support by the Deutsche Bundesstiftung Umwelt (DBU) through the PhD scholarship programme. EB acknowledges funding from the Alexander von Humboldt Sofja Kovalevskaja Award.

We thank Frauke Nevoigt, Ute Hecht und Karl Bumke for providing the temperature data set. We also thank Rainer Kiko for handling temperature data sets and for providing ZooScan. We thank Frank Melzner for intellectual input on the heat wave design. We also thank all research divers for species collections. Renate Schütt, Philipp Neitzel and Anna-Lena Kolze identified the species.

We also thank the following persons for practical support during the experiment (alphabetical order): Miriam Beck, Sönke Beween, Patrick Blaß, Dennis Brennecke, Isabel Casties, Ricarda Christ, Moritz Ehrlich, Julia Gocks, Chi Guan, Giannina Hattich, Kosmas Hensch, Stefanie Ismar, Ulrike Jäger, Sarah Kaehlert, Clarissa Karthäuser, Mandy Kierspel, Joana Kollert, Tim Lachnit, Serra Örey, Fin-Ole Petersen, Julia Raab, Vincent Saderne, Lara Schmittmann, Sabrina Schneider, Jennifer Schulze, Nadja Stärck, Angela Stippkugel, Jenny Tamm, Fabian Wendt, and Levin Wieschermann for help during the experiment.

AUTHOR CONTRIBUTIONS

CP and MW coordinated the project. CP, MSc and MI collected most of the data. CP coordinated the data assembling and the writing of the manuscript. MSc, MW and FRB analysed the data. CP, MSc, MW and FRB wrote the manuscript. All coauthors contributed to the design of the study, performed the experiments, were involved in the final sampling and contributed to the final draft of the manuscript.

ORCID

Christian Pansch  <http://orcid.org/0000-0001-8442-4502>

Marco Scotti  <http://orcid.org/0000-0002-0775-6148>

Elizabeta Briski  <http://orcid.org/0000-0003-1896-3860>

Florian Weinberger  <http://orcid.org/0000-0003-3366-6880>

REFERENCES

- Ainsworth, T. D., Heron, S. F., Ortiz, J. C., Mumby, P. J., Grech, A., Ogawa, D., ... Leggat, W. (2016). Climate change disables coral bleaching protection on the Great Barrier Reef. *Science*, 352, 338–342. <https://doi.org/10.1126/science.aac7125>
- Allison, G. (2004). The influence of species diversity and stress intensity on community resistance and resilience. *Ecological Monographs*, 74, 117–134. <https://doi.org/10.1890/02-0681>
- Bates, D., Maechler, M., Bolker, B., & Walker, S. (2017). Linear Mixed-Effects Models using 'Eigen' and S4. R package version 1.1-13. <http://cran.r-project.org/package=lme4>
- Benedetti-Cecchi, L., Bertocci, I., Vaselli, S., & Maggi, E. (2006). Temporal variance reverses the impact of high mean intensity of stress in climate change experiments. *Ecology*, 87, 2489–2499. [https://doi.org/10.1890/0012-9658\(2006\)87\[2489:TVRTIO\]2.0.CO;2](https://doi.org/10.1890/0012-9658(2006)87[2489:TVRTIO]2.0.CO;2)
- Beukema, J. J., Dekker, R., & Jansen, J. M. (2009). Some like it cold: Populations of the tellinid bivalve *Macoma balthica* (L.) suffer in various ways from a warming climate. *Marine Ecology Progress Series*, 384, 135–145. <https://doi.org/10.3354/meps07952>
- Blake, J., & Maciolek, N. (1987). A redescription of *Polydora cornuta* (Polychaeta: Spionidae) and designation of a neotype. *Bulletin of the Biological Society of Washington*, 7, 11–15.
- Bock, M. J., & Miller, D. C. (1996). Fluid flow and suspended particulates as determinants of polychaete feeding behavior. *Journal of Marine Research*, 54, 565–588. <https://doi.org/10.1357/0022240963213547>
- Bonsdorff, E., Norkko, A., & Boström, C. (1995). Recruitment and population maintenance of the bivalve *Macoma balthica* (L.) - factors affecting settling success and early survival on shallow sandy bottoms. Proceedings of the 28th European Marine Biological Symposium. Fredensborg, Olsen and Olsen.
- Boyd, P. W., Cornwall, C. E., Davison, A., Doney, S. C., Fourquez, M., Hurd, C. L., ... McMin, A. (2016). Biological responses to environmental heterogeneity under future ocean conditions. *Global Change Biology*, 22, 2633–2650. <https://doi.org/10.1111/gcb.13287>
- Bozinovic, F., Bastias, D. A., Boher, F., Clavijo-Baquet, S., Estay, S. A., & Angilletta, M. J. (2011). The mean and variance of environmental temperature interact to determine physiological tolerance and fitness. *Physiological and Biochemical Zoology*, 84, 543–552. <https://doi.org/10.1086/662551>
- Chaparron, C., & Seuront, L. (2011). Behavioral thermoregulation in a tropical gastropod: Links to climate change scenarios. *Global Change Biology*, 17, 1740–1749. <https://doi.org/10.1111/j.1365-2486.2010.02356.x>
- Cloern, J. E., Abreu, P. C., Carstensen, J., Chauvaud, L., Elmgren, R., Grall, J., ... Yin, K. (2016). Human activities and climate variability drive fast-paced change across the world's estuarine-coastal ecosystems. *Global Change Biology*, 22, 513–529. <https://doi.org/10.1111/gcb.13059>
- Costanza, R., Darge, R., Degroot, R., Farber, S., Grasso, M., Hannon, B., ... van den Belt, M. (1997). The value of the world's ecosystem services and natural capital. *Nature*, 387, 253–260. <https://doi.org/10.1038/387253a0>
- Dauer, D. M., Maybury, C. A., & Ewing, R. M. (1981). Feeding-behavior and general ecology of several spionid polychaetes from the Chesapeake bay. *Journal of Experimental Marine Biology and Ecology*, 54, 21–38. [https://doi.org/10.1016/0022-0981\(81\)90100-3](https://doi.org/10.1016/0022-0981(81)90100-3)
- Daufresne, M., Bady, P., & Fruget, J. F. (2007). Impacts of global changes and extreme hydroclimatic events on macroinvertebrate community structures in the French Rhone River. *Oecologia*, 151, 544–559. <https://doi.org/10.1007/s00442-006-0655-1>
- Delgado, L., Guerao, G., & Ribera, C. (2011). Effects of different salinities on juvenile growth of *Gammarus aequicauda* (Malacostraca: Amphipoda). *International Journal of Zoology*, 2011, 6 248790.
- Estay, S. A., Clavijo-Baquet, S., Lima, M., & Bozinovic, F. (2011). Beyond average: An experimental test of temperature variability on the population dynamics of *Tribolium confusum*. *Population Ecology*, 53, 53–58. <https://doi.org/10.1007/s10144-010-0216-7>
- Garrabou, J., Coma, R., Bensoussan, N., Bally, M., Chevaldonné, P., Cigliano, M., ... Cerrano, C. (2009). Mass mortality in Northwestern Mediterranean rocky benthic communities: Effects of the 2003 heat wave. *Global Change Biology*, 15, 1090–1103. <https://doi.org/10.1111/j.1365-2486.2008.01823.x>
- Gerdol, V., & Hughes, R. G. (1994a). Effect of *Corophium volutator* on the abundance of benthic diatoms, bacteria and sediment stability, in 2 Estuaries in Southeastern England. *Marine Ecology Progress Series*, 114, 109–115. <https://doi.org/10.3354/meps114109>
- Gerdol, V., & Hughes, R. G. (1994b). Feeding behavior and diet of *Corophium volutator* in an estuary in Southeastern England. *Marine Ecology Progress Series*, 114, 103–108. <https://doi.org/10.3354/meps114103>
- Glynn, P. W., & Dcroz, L. (1990). Experimental evidence for high-temperature stress as the cause of el-nino-coincident coral mortality. *Coral Reefs*, 8, 181–191. <https://doi.org/10.1007/BF00265009>
- Goldenberg, S. U., Nagelkerken, I., Marangon, E., Bonnet, A., Ferreira, C. M., & Connell, S. D. (2018). Ecological complexity buffers the impacts of future climate on marine consumers. *Nature Climate Change*, 8, 229–233. <https://doi.org/10.1038/s41558-018-0086-0>
- Gorsky, G., Ohman, M. D., Picheral, M., Gasparini, S., Stemann, L., Romagnan, J.-B., ... Prejger, F. (2010). Digital zooplankton image analysis using the ZooScan integrated system. *Journal of Plankton Research*, 32, 285–303. <https://doi.org/10.1093/plankt/fbp124>
- Gräwe, U., Friedland, R., & Burchard, H. (2013). The future of the western Baltic Sea: Two possible scenarios. *Ocean Dynamics*, 63, 901–921. <https://doi.org/10.1007/s10236-013-0634-0>
- Griffiths, J. R., Kadin, M., Nascimento, F. J. A., Tamelander, T., Törnroos, A., Bonaglia, S., ... Winder, M. (2017). The importance of benthic-pelagic coupling for marine ecosystem functioning in a changing world. *Global Change Biology*, 23, 2179–2196. <https://doi.org/10.1111/gcb.13642>
- Gülzow, E. (2015). Studies on the influence of temperature on the feeding rates of important mesograzers in the western Baltic Sea. Unpublished Masters of Science Carl von Ossietzky Universität Oldenburg, Carl von Ossietzky Universität Oldenburg.
- Gunderson, A. R., Armstrong, E. J., & Stillman, J. H. (2016). Multiple stressors in a changing world: The need for an improved perspective on physiological responses to the dynamic marine environment. *Annual Review of Marine Science*, 8(8), 357–378. <https://doi.org/10.1146/annurev-marine-122414-033953>
- Hagerthey, S. E., Defew, E. C., & Paterson, D. M. (2002). Influence of *Corophium volutator* and *Hydrobia ulvae* on intertidal benthic diatom assemblages under different nutrient and temperature regimes. *Marine Ecology Progress Series*, 245, 47–59. <https://doi.org/10.3354/meps245047>
- Halpern, B. S., Walbridge, S., Selkoe, K. A., Kappel, C. V., Micheli, F., D'Agrosa, C., ... Watson, R. (2008). A global map of human impact on marine ecosystems. *Science*, 319, 948–952. <https://doi.org/10.1126/science.1149345>
- Harley, C. D. G. (2008). Tidal dynamics, topographic orientation, and temperature-mediated mass mortalities on rocky shores. *Marine Ecology Progress Series*, 371, 37–46. <https://doi.org/10.3354/meps07711>
- Harris, M. P., Beare, D., Toresen, R., Nøttestad, L., Kloppmann, M., Dörner, H., ... Wanless, S. (2007). A major increase in snake pipefish (*Entelurus aequoreus*) in northern European seas since 2003: Potential implications for seabird breeding success. *Marine Biology*, 151, 973–983. <https://doi.org/10.1007/s00227-006-0534-7>
- Helmuth, B., Russell, B. D., Connell, S. D., Dong, Y., Harley, C. D. G., Lima, F. P., ... Mieszkowska, N. (2014). Beyond long-term averages: Making biological sense of a rapidly changing world. *Climate Change Responses*, 1, 6. <https://doi.org/10.1186/s40665-014-0006-0>
- Hobday, A. J., Alexander, L. V., Perkins, S. E., Smale, D. A., Straub, S. C., Oliver, E. C. J., ... Wernberg, T. (2016). A hierarchical approach to

- defining marine heatwaves. *Progress in Oceanography*, 141, 227–238. <https://doi.org/10.1016/j.pocean.2015.12.014>
- Holland, A. F., Mountford, N. K., & Mihursky, J. A. (1977). Temporal variation in upper bay mesohaline benthic communities: I. The 9m mud habitat. *Chesapeake Science*, 18, 370–378. <https://doi.org/10.2307/1350592>
- Huey, R. B., Carlson, M., Crozier, L., Frazier, M., Hamilton, H., Harley, C., ... Kingsolver, J. G. (2002). Plants versus animals: Do they deal with stress in different ways? *Integrative and Comparative Biology*, 42, 415–423. <https://doi.org/10.1093/icb/42.3.415>
- IPCC. (2013). Climate Change 2013: The Physical Science Basis. In T. F. Stocker, D. Qin, G.-K. Plattner, M. Tignor, S. K. Allen, J. Boschung, A. Nauels, Y. Xia, V. Bex & P. M. Midgley (Eds.), *Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. New York, NY: Cambridge University Press, 1535 pp.
- Jentsch, A., Kreyling, J., & Beierkuhnlein, C. (2007). A new generation of climate-change experiments: Events, not trends. *Frontiers in Ecology and the Environment*, 5, 365–374. [https://doi.org/10.1890/1540-9295\(2007\)5\[365:ANGOCE\]2.0.CO;2](https://doi.org/10.1890/1540-9295(2007)5[365:ANGOCE]2.0.CO;2)
- Johannesson, K., & Andre, C. (2006). Life on the margin: Genetic isolation and diversity loss in a peripheral marine ecosystem, the Baltic Sea. *Molecular Ecology*, 15, 2013–2029. <https://doi.org/10.1111/j.1365-294X.2006.02919.x>
- Johnk, K. D., Huisman, J., Sharples, J., Sommeijer, B., Visser, P. M., & Stroom, J. M. (2008). Summer heatwaves promote blooms of harmful cyanobacteria. *Global Change Biology*, 14, 495–512. <https://doi.org/10.1111/j.1365-2486.2007.01510.x>
- Laine, A. O. (2003). Distribution of soft-bottom macrofauna in the deep open Baltic Sea in relation to environmental variability. *Estuarine Coastal and Shelf Science*, 57, 87–97. [https://doi.org/10.1016/S0272-7714\(02\)00333-5](https://doi.org/10.1016/S0272-7714(02)00333-5)
- Long, W. C., Seitz, R. D., Brylawski, B. J., & Lipcius, R. N. (2014). Individual, population, and ecosystem effects of hypoxia on a dominant benthic bivalve in Chesapeake Bay. *Ecological Monographs*, 84, 303–327. <https://doi.org/10.1890/13-0440.1>
- Lotze, H. K., Lenihan, H. S., Bourque, B. J., Bradbury, R. H., Cooke, R. G., Kay, M. C., ... Jackson, J. B. (2006). Depletion, degradation, and recovery potential of estuaries and coastal seas. *Science*, 312, 1806–1809. <https://doi.org/10.1126/science.1128035>
- Marbà, N., Jorda, G., Agusti, S., Girard, C., & Duarte, C. M. (2015). Footprints of climate change on Mediterranean Sea biota. *Frontiers in Marine Science*, 2, 56.
- Meadows, P. S., & Ruagh, A. A. (1981). Temperature preferences and activity of *Corophium-Volutator* (Pallas) in a new choice apparatus. *Sarsia*, 66, 67–72. <https://doi.org/10.1080/00364827.1981.10414522>
- Meehl, G. A., Karl, T., Easterling, D. R., Changnon, S., Pielke, R., Changnon, D., ... Zwiers, F. (2000). An introduction to trends in extreme weather and climate events: Observations, socioeconomic impacts, terrestrial ecological impacts, and model projections. *Bulletin of the American Meteorological Society*, 81, 413–416. [https://doi.org/10.1175/1520-0477\(2000\)081<0413:AITTIE>2.3.CO;2](https://doi.org/10.1175/1520-0477(2000)081<0413:AITTIE>2.3.CO;2)
- Meehl, G. A., & Tebaldi, C. (2004). More intense, more frequent, and longer lasting heat waves in the 21st century. *Science*, 305, 994–997. <https://doi.org/10.1126/science.1098704>
- Melzner, F., Thomsen, J., Koeve, W., Oschlies, A., Gutowska, M. A., Bange, H. W., ... Körtzinger, A. (2012). Future ocean acidification will be amplified by hypoxia in coastal habitats. *Marine Biology*, 160, 1875–1888.
- Micheli, F., & Halpern, B. S. (2005). Low functional redundancy in coastal marine assemblages. *Ecology Letters*, 8, 391–400. <https://doi.org/10.1111/j.1461-0248.2005.00731.x>
- Mouthon, J., & Daufresne, M. (2006). Effects of the 2003 heatwave and climatic warming on mollusc communities of the Saône: A large lowland river and of its two main tributaries (France). *Global Change Biology*, 12, 441–449. <https://doi.org/10.1111/j.1365-2486.2006.01095.x>
- Paaijmans, K. P., Heinig, R. L., Seliga, R. A., Blanford, J. I., Blanford, S., Murdock, C. C., & Thomas, M. B. (2013). Temperature variation makes ectotherms more sensitive to climate change. *Global Change Biology*, 19, 2373–2380. <https://doi.org/10.1111/gcb.12240>
- Pansch, C., Scotti, M., Barboza, F. R., Al-Janabi, B., Brakel, J., Briski, E., ... Wahl, M. (2018). Heat waves and their significance for a temperate benthic community: a near-natural experimental approach. *PANGAEA*. <https://doi.org/10.1594/PANGAEA.888578>
- Parmesan, C., Root, T. L., & Willig, M. R. (2000). Impacts of extreme weather and climate on terrestrial biota. *Bulletin of the American Meteorological Society*, 81, 443–450. [https://doi.org/10.1175/1520-0477\(2000\)081<0443:IOEWAC>2.3.CO;2](https://doi.org/10.1175/1520-0477(2000)081<0443:IOEWAC>2.3.CO;2)
- Rahmstorf, S., & Coumou, D. (2012). Increase of extreme events in a warming world (vol 108, pg 17905, 2011). *Proceedings of the National Academy of Sciences of the United States of America*, 108, 17905–17909.
- Reusch, T. B. H., Ehlers, A., Hammerli, A., & Worm, B. (2005). Ecosystem recovery after climatic extremes enhanced by genotypic diversity. *Proceedings of the National Academy of Sciences of the United States of America*, 102, 2826–2831. <https://doi.org/10.1073/pnas.0500008102>
- Riebesell, U., & Gattuso, J.-P. (2015). Lessons learned from ocean acidification research. *Nature Climate Change*, 5, 3.
- Rumohr, H., Brey, T., & Ankar, S. (1987). *A compilation of biometric conversion factors for benthic invertebrates of the Baltic Sea*. Kiel, Germany: Institut für Meereskunde.
- Scannell, H. A., Pershing, A. J., Alexander, M. A., Thomas, A. C., & Mills, K. E. (2016). Frequency of marine heatwaves in the North Atlantic and North Pacific since 1950. *Geophysical Research Letters*, 43, 2069–2076. <https://doi.org/10.1002/2015GL067308>
- Schild, L. C., & Glauser, D. A. (2013). Dynamic switching between escape and avoidance regimes reduces *Caenorhabditis elegans* exposure to noxious heat. *Nature Communications*, 4, 1–11.
- Shimeta, J., Witucki, P. F., & Hippe, K. R. (2004). Influences of nutritional state and temperature on suspension-feeding rates and mechanics in the spionid polychaete *Polydora cornuta*. *Marine Ecology Progress Series*, 280, 173–180. <https://doi.org/10.3354/meps280173>
- Skadsheim, A. (1990). A cohort life table for *gammarus-salinus* (Amphipoda). *Oikos*, 57, 207–214. <https://doi.org/10.2307/3565941>
- Sokolova, I. M., Frederich, M., Bagwe, R., Lannig, G., & Sukhotin, A. A. (2012). Energy homeostasis as an integrative tool for assessing limits of environmental stress tolerance in aquatic invertebrates. *Marine Environmental Research*, 79, 1–15. <https://doi.org/10.1016/j.marenvres.2012.04.003>
- Sorte, C. J. B., Fuller, A., & Bracken, M. E. S. (2010). Impacts of a simulated heat wave on composition of a marine community. *Oikos*, 119, 1909–1918. <https://doi.org/10.1111/j.1600-0706.2010.18663.x>
- Stewart, R. I. A., Dossena, M., Bohan, D. A., Jeppesen, E., Kordas, R. L., Ledger, M. E., ... Woodward, G. (2013). Mesocosm experiments as a tool for ecological climate-change research. *Advances in Ecological Research*, 48, 71–181. <https://doi.org/10.1016/B978-0-12-417199-2.00002-1>
- Swaty, R. L., Deckert, R. J., Whitham, T. G., & Gehring, C. A. (2004). Ectomycorrhizal abundance and community composition shifts with drought: Predictions from tree rings. *Ecology*, 85, 1072–1084. <https://doi.org/10.1890/03-0224>
- Taghon, G. L., Nowell, A. R. M., & Jumars, P. A. (1980). Induction of suspension feeding in spionid polychaetes by high particulate fluxes. *Science*, 210, 562–564. <https://doi.org/10.1126/science.210.4469.562>
- Van Der Veer, H. W., Cardoso, J. F. M. F., & Van Der Meer, J. (2006). The estimation of DEB parameters for various Northeast Atlantic bivalve species. *Journal of Sea Research*, 56, 107–124. <https://doi.org/10.1016/j.seares.2006.03.005>

- Vasseur, D. A., Delong, J. P., Gilbert, B., Greig, H. S., Harley, C. D., McCann, K. S., ... O'Connor, M. I. (2014). Increased temperature variation poses a greater risk to species than climate warming. *Proceedings of the Royal Society of London, Series B: Biological Sciences*, 281, 20132612. <https://doi.org/10.1098/rspb.2013.2612>
- Wahl, M. (2009). *Marine hard bottom communities: Patterns, dynamics, diversity and change*. Heidelberg, Germany: Springer. <https://doi.org/10.1007/b76710>
- Wahl, M., Buchholz, B., Winde, V., Golomb, D., Guy-Haim, T., Müller, J., ... Böttcher, M. E. (2015). A mesocosm concept for the simulation of near-natural shallow underwater climates: The Kiel Outdoor Benthocosms (KOB). *Limnology and Oceanography-Methods*, 13, 651–663. <https://doi.org/10.1002/lom3.10055>
- Wahl, M., Saderne, V., & Sawall, Y. (2016). How good are we at assessing the impact of ocean acidification in coastal systems? Limitations, omissions and strengths of commonly used experimental approaches with special emphasis on the neglected role of fluctuations. *Marine and Freshwater Research*, 67, 25–36. <https://doi.org/10.1071/MF14154>
- Walter, J., Jentsch, A., Beierkuhnlein, C., & Kreyling, J. (2013). Ecological stress memory and cross stress tolerance in plants in the face of climate extremes. *Environmental and Experimental Botany*, 94, 3–8. <https://doi.org/10.1016/j.envexpbot.2012.02.009>
- Weinberger, F., Rohde, S., Oschmann, Y., Shahnaz, L., Dobretsov, S., & Wahl, M. (2011). Effects of limitation stress and of disruptive stress on induced antigrazing defense in the bladder wrack *Fucus vesiculosus*. *Marine Ecology-Progress Series*, 427, 83–94. <https://doi.org/10.3354/meps09044>
- Wernberg, T., Smale, D. A., & Thomsen, M. S. (2012). A decade of climate change experiments on marine organisms: Procedures, patterns and problems. *Global Change Biology*, 18, 1491–1498. <https://doi.org/10.1111/j.1365-2486.2012.02656.x>
- Wernberg, T., Smale, D. A., Tuya, F., Thomsen, M. S., de Langlois, T. J., Bettignies, T., ... Rousseaux, C. S. (2013). An extreme climatic event alters marine ecosystem structure in a global biodiversity hotspot. *Nature Climate Change*, 3, 78–82. <https://doi.org/10.1038/nclimate1627>
- Werner, F. J., Graiff, A., & Matthiessen, B. (2016). Temperature effects on seaweed-sustaining top-down control vary with season. *Oecologia*, 180, 889–901. <https://doi.org/10.1007/s00442-015-3489-x>
- Wood, S. N. (2004). Stable and efficient multiple smoothing parameter estimation for generalized additive models. *Journal of the American Statistical Association*, 99, 673–686. <https://doi.org/10.1198/016214504000000980>
- Wood, S. N. (2006). *Generalized additive models: An introduction with R*. Boca Raton, Florida: CRC Press Taylor and Francis Group - A Chapman and Hall Book.
- Woodin, S. A., Hilbish, T. J., Helmuth, B., Jones, S. J., & Wetthey, D. S. (2013). Climate change, species distribution models, and physiological performance metrics: Predicting when biogeographic models are likely to fail. *Ecology and Evolution*, 3, 3334–3346.
- Zajac, R. N. (1991a). Population ecology of polydora-ligni (Polychaeta, Spionidae). 1. Seasonal-variation in population characteristics and reproductive activity. *Marine Ecology Progress Series*, 77, 197–206. <https://doi.org/10.3354/meps077197>
- Zajac, R. N. (1991b). Population ecology of Polydora ligni (Polychaeta, Spionidae). 2. Seasonal demographic variation and its potential impact on life-history evolution. *Marine Ecology Progress Series*, 77, 207–220. <https://doi.org/10.3354/meps077207>
- Zajac, R. N., & Whitlatch, R. B. (1982). Responses of estuarine Infauna to disturbance. 2. Spatial and temporal variation of succession. *Marine Ecology Progress Series*, 10, 15–27. <https://doi.org/10.3354/meps010015>
- Zenetos, A., Gofas, S., Morri, C., Rosso, A., Violanti, D., Garcia Raso, J. E., ... Verlaque, M. (2012). Alien species in the Mediterranean Sea by 2012. A contribution to the application of European Union's Marine Strategy Framework Directive (MSFD). Part 2. Introduction trends and pathways. *Mediterranean Marine Science*, 13, 328–352. <https://doi.org/10.12681/mms.327>

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Pansch C, Scotti M, Barboza FR, et al. Heat waves and their significance for a temperate benthic community: A near-natural experimental approach. *Glob Change Biol*. 2018;24:4357–4367. <https://doi.org/10.1111/gcb.14282>